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Chapter 27
Chips Aren't Cash: Tournaments, Part II

A Survey of Equity Formulas

The Theory of Doubling Up is not designed to capture situations near the money or at the final table. Yet we commonly want to have an explicit mapping of our tournament chip stack to its cash value is near the conclusion of a tournament, either because there is the possibility of dealmaking (when this is allowed) or to account for the payout structure's distortion of the chip to cash mapping so that we can make more accurate decisions.

In fact, it is worth noting that the two main sources of this distortion (skill and the payout structure) each apply at different moments throughout the tournament. At the beginning of the tournament, the payout structure is of relatively small importance, but skill is at its maximum effect. By contrast, at the end of a tournament, the blinds are often high and so skill is fairly reduced (although the specific skill of playing shorthanded might be emphasized), while the payout structure's effect on the mapping is maximized.

In any event, there is no easy answer to the question of mapping chip stacks to prize equity, even for players of equal skill. Various approaches have been proposed to generally solve this problem. In this section, we present a number of different models that have been proposed and make some comments about the biases which are present in each. We begin by defining some terms.

Each of the i players remaining in a tournament has a chip stack S_i. The ***total number of chips remaining*** in the tournament $\mathcal{T}$ is then:

$$\mathcal{T} = \sum S_i$$

A player's fraction of the remaining chips is:

$$s_i = \frac{S_i}{\mathcal{T}}$$

Obviously the sum of all s_i values will be 1.

Additionally, we define v_j as the prize payout for finishing in j^{th} place. The sum of all of the prizes is the total prize pool V. The ***prize pool remaining to be paid out*** when there are n players left V_n is as follows:

$$V_n = \sum_{k=1}^{n} v_k$$

We can also consider the ***incremental prize pool,*** the amount left to be decided by players finishing in places other than nth place: Letting w_j be the incremental prize pool for j^{th} place, and W_n the cumulative prize pool through n^{th} place:

$$w_j = v_j - v_n$$

$$W_n = \sum_{1}^{n} w_j = V_n - nv_n$$

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Since the v_n value is a constant, the decision processes within the tournament for the ***gross prize pool payouts*** and the ***incremental prize pool payouts*** are identical.

A tournament is considered ***in the money*** if $v_n > 0$; that is, the next player to bust receives a payout. This occurs at the precise n value where $v_n = 0$ but $v_n + 1 > 0$, where the tournament is said to be ***on the bubble.***

Now we define $p_{i,j}$ as the probability that player *i* finishes in place *j*. Then we have a basic prize equity formula:

$$< X_i >= \sum_{j=1,N} p_{i,j} v_j$$

Most equity formulas that have been presented assume that the chance of finishing in first place $p_{i,1}$ is approximately equal to the fraction of chips s_i. This is true if the players are of equal skill; however, in situations where the players are of unequal skill, methods such as the Theory of Doubling Up could be used to estimate the chance of finishing first. Instead of using chip stacks as a proxy for the chance of finishing first, we will simply define x_i as the probability of player *i* finishing first. This can be substituted with any probability-generating method you wish.

Equity formulas typically take the chance of finishing in first place as a given, and provide a simplified methodology for assigning probabilities to the other places.

The Proportional Chip Count Formula

The first model for assigning probabilities to the other places that we present is the frequently employed "chip count" method, which can be expressed as follows:

$$<X_i >= v_n + s_i W_n \qquad \textbf{(27.1)}$$

This model consists of two parts:

- First, all players receive the guaranteed value of the finish that they are guaranteed.
- Then, the remaining prize pool is divided amongst the players according to then chip count.

For example, suppose the stacks are 25,000, 20,000, 10,000, and 5,000, and the payouts are \$2,000, \$1,400, \$900, and \$700, Then the "chip count" method using Equation 27.1 would allocate the remaining prize pool of \$5,000 as follows:

\$2,800 is distributed equally to each player as his guaranteed payout.

The remaining \$2,600 divided to each player according to his chip count.

Player 1: $<X_1> = 700 + (5/12)(2200) = \$1,616.67$
Player 2: $<X_2> = 700 + (1/3)(2200) = \$1,433.33$
Player 3: $<X_3> = 700 + (1/6)(2200) = \$1,066.67$
Player 4: $<X_4> = 700 + (1/12)(2200) = \$883.33$

At a glance, perhaps these numbers look reasonable. But the flaw of this method can quickly be seen if we change the stacks so that Player 1 has 48,000 and the other players 4,000 each:

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Player 1: $\langle X_1 \rangle = 700 + (^4/_5)(2200) = \$2,460$
Players 2-4: $\langle X_i \rangle = 700 + (^1/_{15})\{2200) = \$846.67$

Clearly this isn't right. Player 1 seems to be entitled to receive more than first place money! This is a systemic flaw in this method; it tends to distribute too much of the prize pool to players on the basis of their chips, and too little to players on the basis of their chairs. The more equal the stacks are, the more accurate this formula is. However, this is true of all models - any model that does not tend to equalize the payouts as the x_i values equalize cannot be right!

The reason for this flaw is that this model is equivalent to saying that $p_{i,j} = s_i$ for all values of *j!* Well, we can see how this is wrong. Suppose our player is the chip leader (equally skilled). Then his chance of 1st place is larger than that of all the other players. But his chance of 2nd, 3rd, 4th and so on down to nth is the same. So his total chance of finishing in any place in the tournament is ns_i. If s_i is larger than $1/n$, then his total chance of finishing in some place in the tournament will be higher than 1. This is obviously incorrect.

One potential modification we can make to this formula is to constrain the finish data to sum to one by making the following corrections:

$$m = \left\lfloor \frac{1}{x_i} \right\rfloor$$
$$p_{i,j \le m} = x_i$$
$$p_{i,m+1} = 1 - m\, x_i$$

Here, the brackets in the expression for m are the ***floor function,*** which returns the largest integer less than or equal to the bracketed expression.

Hence, if $x_i > 0.5$, then $m = 1$, and we have

$$p_{i,1} = x_i$$
$$p_{i,2} = 1 - x_i$$

This approximation is a more reasonable one when the stacks are very large. Consider the revised probabilities for the example we just considered:

For Player 1, who has $^5/_{12}$ of the chips, $m = 2$.

Then

$$p_{i,1} = {}^5/_{12}$$
$$p_{i,2} = {}^5/_{12}$$
$$p_{i,3} = {}^1/_{16}$$

and l's revised equity is:

$$\langle X_1 \rangle = 700 + (^5/_{12})(1300) + (^5/_{12})(700) + (^1/_6)(200) = \$1,566.67$$

Unfortunately, this modification causes the formula to no longer sum to the total prize pool. Therefore, it is primarily useful to establish a slightly lower upper bound for the equities of the chip leaders. It is still fairly biased on behalf of the tall stacks, however.

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The Landrum-Burns Formula

The second model we will consider was originally proposed by Stephen Landrum and Jazbo Burns on rec.gambling.poker. This model uses the following methodology:

We calculate each player's chance of finishing first.
We then assume that when he does not finish first, he finishes equally often in the remaining places.

$$< X_i >= p_{i,1}v_1 + \sum_{k=2}^{n} \frac{(1-p_{i,1})}{n-1} v_k \qquad \textbf{(27.2)}$$

Using this formula, we can recalculate our previous example where the stacks are 25,000, 20,000, 10,000, and 5,000, and the payouts are $2,000, $1,400, $900, and $700. Then the "Landrum-Burns" method calculates the equities as follows:

Player 1: $(^5/_{12})(2200) + (^7/_{36})(1400) + (^7/_{36})(900) + (^7/_{36})(700) = \$1{,}416.67$
Player 2: $(^1/_3)(2200) + (^2/_9)(1400) + (^2/_9)(900) + (^2/_9)(700) = \$1{,}333.33$
Player 3: $(^1/_6)(2200) + (^5/_{18})(1400) + (^5/_{18})(900) + (^5/_{18})(700) = \$1{,}066.67$
Player 4: $(^1/_{12})(2200) + (^{11}/_{36})(1400) + (^{11}/_{36})(900) + (^{11}/_{36})(700) = \$1{,}083.33$

But we can equally expose the error of this method by considering the second example, where Player 1 has 48,000 and the other players 4,000:

Player 1: $(^4/_5)(2000) + (^1/_{15})(1400) + (^1/_{15})(900) + (^1/_{15})(700) = \$1{,}800$
Players 2-4: $(^1/_{15})(2000) + (^{14}/_{45})(1400) + (^{14}/_{45})(900) + (^{14}/_{45})(700) = \$1{,}066.67$

The $1,800 number is at least "reasonable," since it doesn't exceed the payout for first (as occurred with the "chip count" method). However, calculating it directly exposes a difficulty with the method. Player 1 may protest that he is not as likely to finish fourth as second, should he fail to win. In fact, he will finish fourth very infrequently, if the blinds are fairly high. So the "fair" price should be adjusted up from the Landrum-Burns value. Likewise, suppose one player has a tiny stack compared to his opponents. He is not as likely to finish second as he is to finish fourth.

This is the bias of the Landrum-Burns method - it overvalues small stacks at the expense of big stacks. This occurs because large stacks will not bust out immediately as often as they finish second. The proportional chip count method and Landrum-Buras together provide us a lower and upper bound to the plausible values; the "true" fair value lies somewhere between them.

In *Gambling Theory and Other Topics,* Mason Malmuth introduced two additional equity formulas.

The Malmuth-Harville Formula

This formula is parallel to the Harville horse-racing formula.
Essentially, what this formula does is the following:

- The chance of finishing first for each player is x_i,
- We can use the following formula to calculate the chance of finishing in 2nd place given that player k has finished first:

$$p(X_{i,2}|X_{k,1}) = \frac{x_i}{1 - x_k}$$

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Then our probability of finishing second is:

$p_{i,2} = p(X_{i,2})$

$$p_{i,2} = \sum_{k \neq i} p\left(X_{i,2} \middle| X_{k,1}\right) p(X_{k,1})$$

$$p_{i,2} = \sum_{i \neq k} \frac{x_i x_k}{1 - x_k}$$

This formula was originally developed for betting favorites in show or place pools in horse racing. However, this formula actually applies more strongly to tournament poker because of an idea pointed out to us by Jeff Yass:

In horse racing, if a horse is a prohibitive favorite, where *x* is closer to one, a substantial amount of his non-wins will be due to a problem or event which will also prevent him from finishing 2nd, such as an injury or disqualification. It is unreasonable, for example, to believe that a horse with an 80% of winning will still have a higher than 80% of finishing second if he fails to win, and even more unreasonable to believe he has a >99% of finishing in the top three.

But in poker tournaments, this effect is minimized. If a tall stack does not win he will still be very likely to finish second, and very- unlikely to finish in the last remaining spot.

We can generalize this rule as well:

$$p(X_{i,j} | X_{1,1} \wedge X_{2,2} \wedge \ldots \wedge X_{j-1,j-1}) = \frac{x_i}{1 - x_1 - \cdots - x_{j-1}} \qquad \textbf{(27.3)}$$

That is, the probability of finishing in the jth place is simply the ratio of winning probability for the player to the win probabilities of the remaining players. In most cases, this formula is a more accurate representation of equity than either the proportional chip formula or Landrum-Burns.

One limitation of this method is that if there are many players remaining it is fairly difficult to calculate at the table because the number of permutations of finishes; as such, it is primarily useful at understanding situations away from the table or for analyzing situations that recur frequently, such as sit-and-go tournament equities or the like.

The Malmuth-Weitzman Formula

A second model proposed by Malmuth in *Gambling Theory and Other Topics* was developed by Mark Weitzman, which we will call the Malmuth-Weitzman formula.

In that text, Malmuth seems to prefer this formula over the Malmuth-Harville formula, although he claims that this formula requires the solution of a system of linear equations for each situation. We believe that it is slightly less accurate than the other formula but the solution to the linear equations is unnecessary, as we shall see.

The main idea of this formula is that when a player busts out of the tournament, his chips are distributed equally (on average) among his opponents.

$$x_i = P\left(X_{i,1}\right) = \sum_{k \neq i} P\left(X_{i,1} \middle| X_{k,n}\right) P(X_{k,n}) = \sum_{k \neq i} \left(x_i + \frac{x_k}{n-1}\right) p_{k,n}$$

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This results in a series of equations that we have to solve. For each player we have a value $p_{i,n}$ which is the probability of busting out next and a summation for all players of the equities from distributing their chips equally among all players.

For example, if we have four players, then we have the following equations (we use p_k here as shorthanded for $p_{k,n}$). All the probabilities shown are of busting out next.

$$x_1 = \left(x_1 + \frac{x_2}{3}\right)p_2 + \left(x_1 + \frac{x_3}{3}\right)p_3 + (x_1 + \frac{x_4}{3})p_4$$
$$x_2 = \left(x_2 + \frac{x_1}{3}\right)p_1 + \left(x_2 + \frac{x_3}{3}\right)p_3 + (x_2 + \frac{x_4}{3})p_4$$
$$x_3 = \left(x_3 + \frac{x_1}{3}\right)p_1 + \left(x_3 + \frac{x_2}{3}\right)p_2 + (x_3 + \frac{x_4}{3})p_4$$

$$x_4 = \left(x_1 + \frac{x_1}{3}\right)p_1 + \left(x_4 + \frac{x_2}{3}\right)p_2 + (x_4 + \frac{x_3}{3})p_3$$

With four equations and four unknowns, we can solve this straightforwardly using algebra. However, we can do better than this. These equations are part of a family of equations called *Vandermonde* equations. We will simply provide the solution to these equations and show that it is correct. We claim that the above matrix implies that the probability of busting out next is inversely proportional to the number of chips (or win probability) left. If we define b as the sum of the inverses of all win probabilities:

$$b = \sum \frac{1}{x_i}$$

Then

$$p_{k,n} = \frac{1/x_k}{b} \qquad \textbf{(27.4)}$$

We can substitute this back into the above formula:

$$x_i = \sum_{k \neq i} (x_i + \frac{x_k}{n-1})p_{k,n}$$

$$x_i = \frac{1}{b}\sum_{k \neq i} (\frac{x_i}{x_k} + \frac{1}{n-1})$$

$$x_i = \frac{1}{b}\sum_{k \neq i} (\frac{x_i}{x_k} + 1)$$

$$x_i = \frac{1}{b}[x_i \sum_{k \neq i} \frac{1}{x_k} + 1]$$

The summation of l/x_k, for all k not equal to I, rather it is $b - 1/x_i$.

So we have:

$$x_i = \frac{1}{b}(bx_i - 1 + 1) = x_i$$

The system of linear equations generated by the assumption of this model implies that each player's chance of going bust next is proportional to the inverse of his chip count or win

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probability.

The Malmuth-Weitzman is a reasonable formula as well and is probably near the edge of some players' ability to calculate at the table, which increases its usefulness. One objection to this formula is that it's a bit sticky when it comes to situations where multiple tables are left in play. If two players are quite short stacked at different tables, it seems counter-intuitive that the player with twice the chips has half the chance of busting. Also, it's a little unclear why players at different tables receive an equal share of the busting chips. However, by and large, this is our favorite formula for assessing tournaments that are in or near the money. When assessing deals at the final table, we use the Malmuth-Weitzman as our baseline equal skill formula.

The important thing to understand from all this is that there are a number of different ways to map chip counts to tournament equities. Because these are all models, they inherently have some sort of bias. Understanding the common methods and the biases they have can lead to an enhanced ability to make in-tournament decisions as well as to make more favorable deals, when the circumstances arise.

Tournament Results and Confidence Intervals

In Part I, we discussed confidence intervals as they applied to win rates. For ring games, confidence intervals can be calculated in a typical fashion. This is because ring games are essentially an aggregation of hand results, which are in turn subject to the law of large numbers. However, for tournaments, constructing confidence intervals is not so easy. Indeed, constructing confidence intervals in the usual way doesn't yield the most accurate analysis.

Consider the case of a winner-take-all tournament with 1000 players. Suppose that we assume that the population's win rates will all fall between -1 and +1 buyin per tournament. Say we play the tournament once. The player who wins the tournament will have experienced a 30 standard deviation event. However, each time the tournament is played, someone is going to win. Even if we play the tournament ten times, it doesn't particularly help matters.

Another difference is that unlike ring games, where estimates of variance quickly converge because of the distribution they follow (chi-squared), we don't normally have good estimates of variance for tournaments. Variance is also highly correlated to win rate because, in tournaments, all large variance events are positive, while negative events are smaller variance (-1 buyin, typically).

The real problem is that for tournaments, it takes many more samples before the distribution converges to a normal distribution.

One problem is skew, which we briefly discussed in Chapter 22. This is defined as:

$$S = \frac{< (X - \mu)^3 >}{\sigma^3}$$

Skew for a n-player winner-take-all tourney is around $\sqrt{n}$, while the skew of a normal distribution is zero. We know the skew of the sum of n identical independent events is the skew of one event divided by $\sqrt{n}$. As we take a larger and larger sample in trials, the skew of the sample will eventually converge on zero. This is one of the ways in which sampling distributions eventually converge on the normal distribution. But here we need more than n samples for the skew to be within 1.

But we can work on the problem nonetheless. The expectation of our gross profit for the

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tournament is:

$$< X >= \sum p_i V_i$$

Where V_i is the prize for finishing in i^{th} place. Let V be the total prize pool and we can define v_i to be the fraction of the prize pool awarded to the ith place finisher.

$$v_i = \frac{V_i}{V}$$

Assuming no rebuys and add-ons, we define the ***size*** of the tournament as the prize pool divided by the number of players n.

$$s = \frac{V}{n}$$

We additionally have the ***cost*** of the tournament b_c, which is the total amount each player must pay to play the tournament. The difference between these two is the ***entry fee***, which can be either positive (when the players pay a fee to the house for spreading the tournament) or negative (such as when the house adds money to the tournament for a guarantee or other promotion). Letting ε_b denote the entry fee:

$$\varepsilon_b = b_c - s$$

For example, suppose we had a tournament that was \$100 + \$10, with a \$50,000 guarantee from the casino. So \$100 from each player's \$110 entry goes to the prize pool; the remainder goes to the house. If the prize poof is short of \$50,000, the casino makes up the difference. If there are 300 players, then we have:

s = \$166.67
b_c = \$110
ε_b = \$56.67

Our net expectation w for entering a tournament is then our expectation of gross profit <X> less the cost of the tournament b_c, or:

$$w = <X> - b_c$$
$$w = \sum p_i V_i - s - \varepsilon_b$$

This expression effectively has two parts. The first part,

$$w_0 = \sum p_i V_i - s$$

should be related to the player's poker and tournament skill, while the second part, ε_b, is be independent of poker skills, except possibly game selection.

We define a ***normalized win rate***, ω, as:

$$\omega \equiv \frac{\omega_0}{s}$$

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$$= \sum p_i \frac{V_i}{s} - 1$$

$$= \sum (np_i) \frac{V_i}{V} - 1$$

$$= \sum (np_i - 1) v_i$$

This normalized win rate is basically the number of buyins (where buyins is normalized to tournament size) that we can expect to net per tournament.

The statistical variable we want to look at is the ***normalized win***, Z, such that:

$$\mathcal{Z} = \frac{X - s}{s}$$

This has the special property that:

$$< Z >= \omega$$

It turns out Z is a statistical variable we are interested in for a number of reasons. The idea here is to get a handle on the behavior of the Z variable, and use that information to construct appropriate confidence intervals.

Note that typically, the mode value for Z is -1.

One possible assumption that would yield an appropriate Z variable is:

$$p_i = \frac{1 + \omega}{n}$$

This assumes that the player's win rate is an effective multiplier on his chance of finishing in each paid place. We begin with the player finishing in each position with equal likelihood. Then we move ω/n finishes from "out of the money" finishes to each payoff spot. This is at odds with the results of the Theory of Doubling Up, which suggests that a skilled player's chance of finishing first is higher than his chance of finishing second, third, and so on. However, we feel this approximation is fairly reasonable to use anyway. Recall from Chapter 25 that:

$$<x^2> = \mu^2 + \sigma^2$$
$$\sigma^2 = <x^2> - \mu^2$$

For this distribution,

$$x = X - b_c$$
$$\mu = < X - b_c >$$

The variance of a single tournament, then, is:

$$\sigma_x^2 =< (X - b_c)^2 > -< X - b_c >^2$$

$$\sigma_x^2 =< X^2 > -< X >^2$$

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$$\sigma_x^2 = \sum p_i V_i^2 - (\omega + b)^2$$

$$\sigma_x^2 = \sum \left(\frac{1+\omega}{n}\right)(Vv_i)^2 - (1+\omega)^2\, b^2$$

$$\sigma_x^2 = \frac{(1+\omega)}{n}(nb)^2 \sum {v_i}^2 - (1+\omega)^2\, b^2$$

$$\sigma_x^2 = \left[n\left(\sum {v_i}^2\right)(1+\omega) - (1+\omega)^2\right] b^2$$

Then we have the variance of Z as:

$$\sigma_z^2 = <\mathcal{Z}^2> - <Z>^2 = \frac{\sigma_x^2}{b^2} = \left(n\sum v_i^2 - 1 - \omega\right)(1+\omega)$$

Hence, the variance of Z is roughly proportional to $1+\omega$ for reasonable values of ω. If we examine the value of:

$$k = n\sum v_i^2 - 1 \qquad \textbf{(27.5)}$$

We will find that this expression is close to the effective number of players that we found when we examined the Theory of Doubling Up. We will call *k* the ***tournament variance multiple.*** Using the variance formula above for the variance of Z is a better estimate of the variance of a tournament than the observed variance of a sample for any sample size that isn't enormous.

Now consider a series of independent tournaments. Suppose that we have a series of normalized outcomes (Z-values) for *m* tournaments.

The average of these normalized outcomes is $\overline{\mathcal{Z}}$ such that:

$$\overline{\mathcal{Z}} = \frac{\sum_{j=1}^{m} \mathcal{Z}_j}{m}$$

Since each $\mathcal{Z}_j$ value is independent of the number of tournaments played, we have the following formula for the variance of a sample of *m* Z-values:

$$\sigma^2 = \frac{\sum \sigma_{zj}^2}{m^2} = \frac{\sum (k_j - \omega)(1+\omega)}{m^2} = \frac{(\overline{k} - \omega)(1+\omega)}{m} \qquad \textbf{(27.6)}$$

In all normal circumstances, ω will be much smaller than *k.* So we can very roughly approximate the above expression with *k/m.* Then to make the variance of a sample of size *m* equal to 1. we must play about the same amount of tournaments as the average variance multiplier for the tournaments. If the variance of the sample is equal to 1, then we have an approximately 68% chance of our observed win rate ω being within 1 buyin of the population mean (neglecting Bayesian implications).

We can call Equation 27.6 the ***win-rate-based variance*** formula, because it uses our observed win rate and the observed structure of the tournament to calculate the variance of the tournament.

This formula has the flaw that in truth, having a win rate of ω does not normally imply that you enhance your chance of finishing in every paying spot by a factor of ω; in fact, some players'

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finishes are more top-heavy than others, with more firsts than seconds and thirds, while others are more weighted toward lower finishes. Another method of calculating this would be to use the Theory of Doubling Up to predict the rate of finish in each spot and calculate the variance that way.

The reason that we do not use the observed variance is that unless the sample size is very large, these formulas will give us a better estimate. If *m* were large however, we could utilize another way of calculating variance, which would be to use the observed win rate $\widehat{\omega}$ and observed variance $\widehat{\sigma^2}$ to calculate *k*.

$$k = \widehat{\omega} + \frac{m\widehat{\sigma^2}}{1 + \widehat{\omega}}$$

We then use this value of *k* in the win-rate-based variance formula. This technique is likely more accurate when *m* is very large, because the convergence of the observed variance to the true variance occurs more rapidly than the convergence of the sampling distribution to the normal distribution.

The question we are trying to answer with confidence intervals is the following:

If our normalized win rate is ω**,** how lucky or unlucky is our observed result?

We cannot strictly use the normal distribution as our guide here because the sampling distribution has not yet converged on the normal distribution. However, we can use the normal distribution to map a ***z-score*** (see Equation 2.6) to a confidence level (if, for example, being more than 2 standard deviations away fails the 95% confidence test) *if* $\omega > \widehat{\omega}$.

$$z = \frac{\widehat{\omega} - \omega}{\sigma(\omega)} < -2$$

If this expression is true, we can conclude that ω fails the test, because we know that the distribution is skew-positive and the probability of having a negative event (having an outcome more than two sigmas below expectation) is actually less than the normal distribution would indicate. So we can exclude some very high win rates based on observed data.

However, for $\omega > \widehat{\omega}$**,** we have to be more careful. The normal distribution is inappropriate for use here because in our sampling distribution, due to the skew, it is more likely that our observed outcome is a positive event than the normal distribution would indicate.

We can, however, use a statistical idea called Tchebyshev's (or Chebyshev's) Rule:

$$P\left(\frac{\widehat{\omega} - \omega}{\sigma(\omega)} = z > a\right) < \frac{1}{a^2}$$

This says that the probability that our outcome is actually in excess of four standard deviations on one side of the mean is always less than $^1/_{16}$. This may seem like a particularly blunt tool, after all, the normal distribution has a similar value of 0.00003. However, this rule has the excellent property of applying to any distribution, no matter how skew.

Let's take a moment now to review all of these ideas in the context of a hypothetical player, player T. Player T is a tournament player, who plays four $50 + $5 tournaments per day. Two of

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the tournaments attract about 100 players (these are the morning tournaments) and the other tournaments attract about 400 players (these are the evening tournaments).

The morning tournaments pay out according to the following schedule:

Place	1	2	3	4	5	6	7	8	9
Pct	30%	20%	12%	10%	8%	6%	5.5%	4.5%	4%

The evening tournaments pay out as follows:

Place	1	2	3	4	5	6	7	8	9	10-18	19-27
Pct	27.5%	17.5%	11.6%	8%	6%	4.5%	3.5%	2.6%	1.7%	1.2%	0.7%

Player T has played these tournaments for 200 days. As a result, he has 400 morning tournament data points, and 400 evening tournament data points.

The following is a list of his finishes in each type of tournament over that span:

Morning tournaments:

Place	**Frequency**	**Gross Buyins**	**Net Buyins**	**Total net**
1st	11	30	29	319
2nd	7	20	19	133
3rd	5	12	11	55
4th	3	10	9	27
5th	8	8	7	56
6th	4	6	5	20
7th	7	5.5	4.5	31.5
8th	2	4.5	3.5	7
9th	5	4	3	15
No cash	348	0	-1	-348
Total	**400**			**315.5**

So Player T's overall cash rate in morning tournaments is $^{52}/_{400}$, or 13%. His overall win is 315.5 buyins, a rate of about 0.79 buyins per tournament.

Evening tournaments:

Place	**Frequency**	**Gross Buyins**	**Net Buyins**	**Total net**
1st	4	110	109	436
2nd	2	70	69	138
3rd	2	46.4	45.4	90.8
4th	0	32	31	0
5th	2	24	23	46
6th	4	18	17	68
7th	1	14	13	13
8th	0	10.4	9.4	0
9th	2	6.8	5.8	11.6
10th - 18th	14	4.8	3.8	53.2

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19th - 27 th	20	2.8	1.8	36
No cash	349	0	-1	-349
Total	400			543.6

And in the evening tournaments, Player T has cashed about 13% of the time again, with an overall win of 543.6 buyins, a rate of about 1.36 buyins per tournament.

Taken together, T has played a total of 800 tournaments, with an aggregate win rate of 1.07 buyins per tournament. We can make an inventory of the statistics we've discussed here:

$\omega = 1.07$
$m = 800$
$\overline{\mathcal{Z}} = 1.07$
$k = n \sum v_i^2 - 1$

For the morning tournaments, we have:

$k = (0.30)^2 + (0.20)^2 + ... + (0.04)^2 = 16.105$

And for the evening tournaments, we have:

$k = (0.275)^2 + (0.175)^2 + ... + (0.007)^2 = 53.6456$

Then our average k value is:

$\overline{k} = 34.875$

and our variance for an 800 tournament sample consisting of half morning and half evening tournaments is approximately:

$$\sigma^2 = \frac{(\overline{k} - \omega)(1 + \omega)}{m} = \frac{(34.875 - 1.07)(1 + 1.75)}{800} = 0.0876$$

or a standard deviation of 0.296 buyins per sample (of 800 tournaments).

The observed sample variance (from the sample actually observed) is about 0.3468 buyins. This difference occurs because the sample is slightly more concentrated in first place finishes. You might object that Player T's apparent tendency to finish first more often than other places skews his underlying distribution. This might be true in practice, although the effect would not be terribly important. However, since we wrote the program that randomly generated the above results, we know that Player T's chance of finishing first was the same as his chance of finishing in any other money position. This is yet another case of the possibility of being fooled by random noise.

Applying the normal distribution to the observed win rate and standard deviation would result in a 95% confidence interval of [0.3802, 1.7675]. This is definitely inaccurate on the high side. Because the distribution is skew-positive, the chance that the true win rate is higher than 1.7675 and T had a very negative event is actually lower than the normal distribution would indicate. On the low side, it is entirely inappropriate to use the normal distribution, because the chance that the observation is actually a positive event from a lower win rate is much higher.

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Utilizing the win rate-based variance calculation, T actually has smaller variance. We find that two standard deviations above would be 1.666 buyins per tournament. This is still slightly inaccurate. In fact, at this point, the sample would fail a hypothesis test for 95% accuracy. The true bound is actually lower than this, but only modestly. On the other side, however, we can use Tchebyshev's Rule and conclude that the chance that the true distribution is lower than 2 standard deviations below the observed mean (or below 0.4818) is no greater than $^1/_4$.

Key Concepts

- A number of equity formulas have been proposed to map chip equities to tournament equities in the late stages of tournaments.
- The Landrum-Burns and Proportional Chip Count methods have clear flaws but are easy to calculate at the table. Since they are flawed in opposite directions, evaluating both can be useful to identify a range of values for the true equity.
- The Malmuth-Harville and Malmuth-Weitznan formulas are more accurate, but harder to implement at the table.
- Tournament equity confidence intervals are an interesting topic because of their significant skew and also because the variance per tournament is highly correlated to the win rate. We can use these facts to obtain more accurate confidence intervals even without strongly significant data (which is often difficult to gather because it takes so many trials).

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